Automated Condition Assessment of Runway Surfaces from Low-Altitude UAV Imagery

Shabaz Ahmad Hasib

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Abstract

Runway pavements wear under aircraft loading, weather and age, and the cracking, vegetation ingress and faded markings that follow are still found mostly by slow, disruptive walking surveys. This paper sets out a vision-based alternative that reads a single low-altitude drone pass and returns a located, prioritised picture of the surface. Frames captured at roughly 1.25 mm per pixel are examined along three independent tracks — a crack-specialised segmentation network and two colour routines for vegetation and painted markings — and every finding is tied to a real distance along the runway by registering successive frames against one another. Over 136 m of surveyed pavement the method maps 545 individual cracks, locates grass growing through the eastern joints, and ranks the worst stretches for follow-up. We report only the quantities the imagery genuinely supports, set them against the ASTM D5340 airfield framework, and are candid about where motion blur and the grooved surface place a ceiling on millimetre-level crack width.

1. Introduction

Runway surface condition is a safety matter as much as a maintenance one. Cracking lets water into the pavement and, once it opens up, sheds fragments that become foreign object debris; vegetation in the joints signals both moisture and neglect; worn markings reduce pilot guidance. International practice (ICAO Annex 14, and CAA CAP 168 in the United Kingdom) expects aerodromes to inspect these surfaces regularly, yet the conventional walking survey is labour-intensive, subjective, and forces runway closures that airports are reluctant to grant.

A drone flying low over the runway offers a way out: it can cover the full length in minutes and record the surface at a resolution fine enough to see individual cracks. The difficulty is turning that imagery into decisions. This paper describes an end-to-end approach that does so, and is deliberately honest about its limits. Our contribution is threefold: a three-track detection design suited to top-down, near-macro imagery; a GPS-free way of placing every finding at a real distance along the runway; and a measured account of which engineering quantities such imagery can, and cannot, be trusted to deliver.

2. Data acquisition

The study uses a single continuous drone pass over an asphalt runway, sampled into 274 frames at five frames per second. The ground sampling distance is about 1.25 mm per pixel — fine enough that a millimetre-scale crack spans several pixels. This places the data in an unusual regime: top-down like aerial survey work, but near-macro like close-range inspection. The runway is grooved for skid resistance, and the longitudinal grooves, together with the slight motion blur of a moving platform, are the two features that most complicate analysis.

3. System architecture

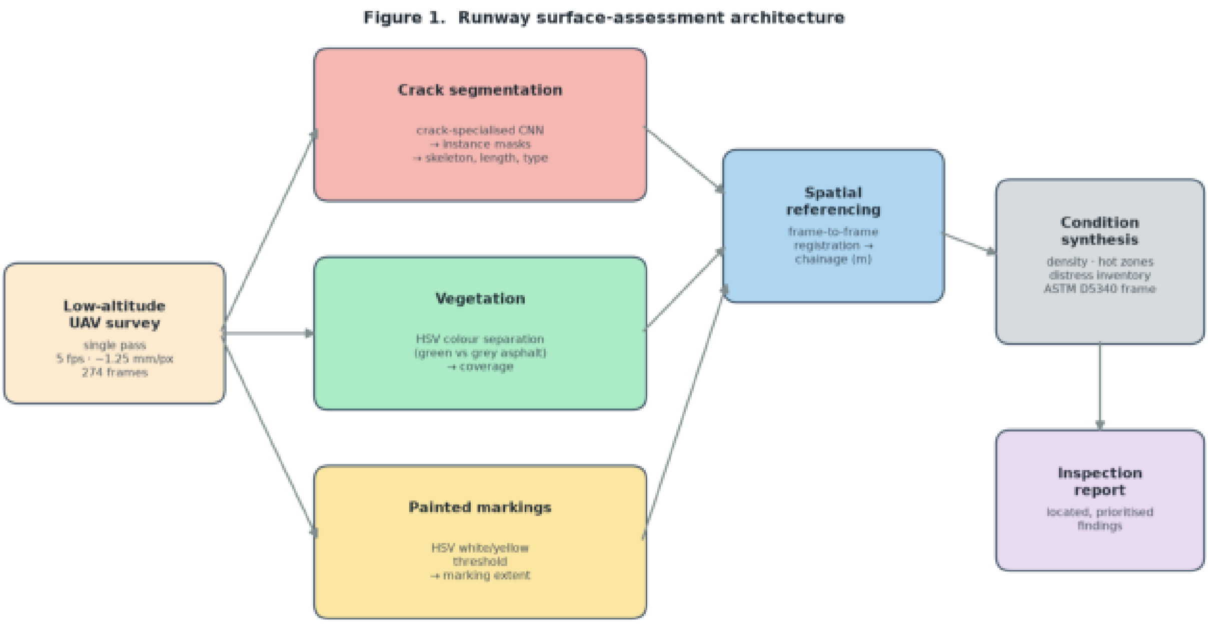

Figure 1. The pipeline reads a single UAV pass and splits the work across three independent detection tracks — crack segmentation, vegetation, and painted markings — before placing each result at a real chainage and drawing the findings together into a prioritised assessment.

Figure 1 shows the overall design. The choice to run three separate tracks, rather than one multi-class detector, is deliberate. The three targets have very different visual signatures and very different data situations: cracks are thin, low-contrast and need a learned model; vegetation and intact paint are strongly coloured and separate cleanly from grey asphalt with simple, transparent colour rules that need no training. Keeping them apart lets each use the most reliable method for its target and keeps the failure modes independent, so a weakness in one track does not corrupt the others.

The first track sends each frame through a crack-specialised convolutional network that outlines cracking as thin shapes rather than coarse boxes, so the area genuinely affected — not the rectangle around it — is what gets measured. The second and third tracks separate green vegetation and bright white or yellow paint from the surrounding asphalt in colour space. A fourth stage recovers how far the platform travelled between consecutive frames and converts frame numbers into metres along the runway. The final stage gathers everything by location into hot zones and a distress inventory, framed against the airfield condition standard.

4. Detection methods

4.1 Surface cracking

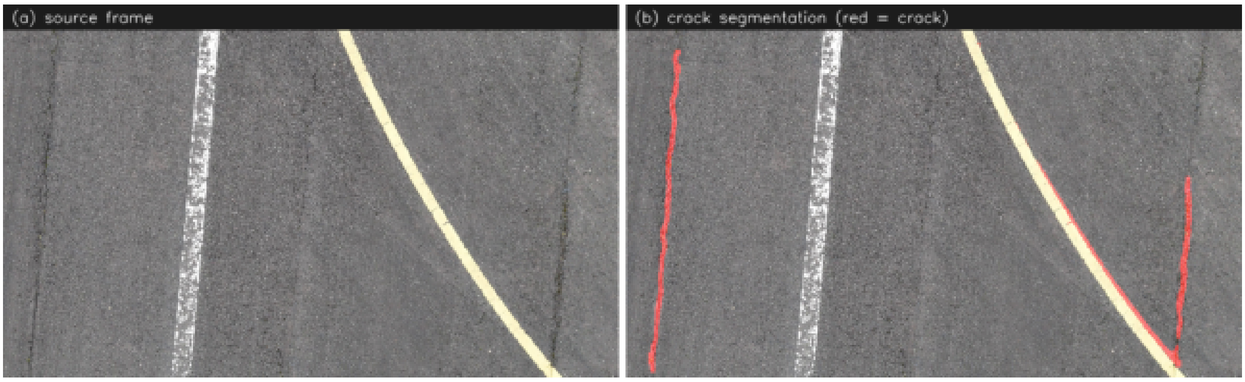

Figure 2. Source frame (left) and the crack outlines returned by the segmentation network (right). The model traces the cracking as thin shapes, which is what allows length and orientation to be measured rather than merely counted.

Cracks are the primary target and the hardest to see. Each frame passes through a segmentation network specialised for cracking (Figure 2), which returns one outline per crack. From each outline we reduce the shape to its centre-line and measure length along that line; the orientation of the line, relative to the runway axis, sorts the crack into longitudinal, transverse, or an interconnected 'alligator' pattern — the last being the one the airfield standard flags for debris risk. Across the survey the network maps 545 cracks totalling roughly 1128 m, dominated by longitudinal cracking (520 cracks) with 18 alligator patches flagged for attention.

4.2 Vegetation encroachment

Figure 3. Grass growing through the joints along the verge is isolated by its colour, which is well separated from grey asphalt and needs no trained model.

Vegetation is the clearest of the three to pick out: living plant matter is green, and green is far from the grey of asphalt in colour space, so a simple hue–saturation rule isolates it reliably (Figure 3). It is also operationally meaningful — grass in the joints points to trapped moisture, is itself a debris source, and is a visible sign that maintenance has lapsed. Vegetation appears in 85 of 274 frames, concentrated along one verge and around the two-thirds mark of the pass, covering about 0.10% of the imaged surface overall.

4.3 Painted markings

Figure 4. Painted lines are picked out by brightness and linear shape, not colour alone. The cream line on the asphalt is traced (magenta) while the grass on the verge — which shares almost the same hue — is correctly left out.

Markings are the subtlest of the three, and colour on its own is misleading: sunlit and dried grass take on almost the same warm hue as faded cream paint, so a plain colour rule flags the verge instead of the line. The track therefore keys on what genuinely sets a marking apart — it is brighter than the asphalt around it, it sits on the grey surface rather than the green verge, and it runs as a long thin line rather than a blob. With those three conditions together the white centre line and the brighter cream lines are traced cleanly and the grass is left alone (Figure 4); markings register in 240 frames. Two cases stay hard: very faded paint, by then barely brighter than the surface beneath it, and the bright concrete shoulder at the runway margin, itself a long pale line — so we read this track as a dependable locator of intact interior markings and a weaker guide to worn ones.

5. Spatial referencing

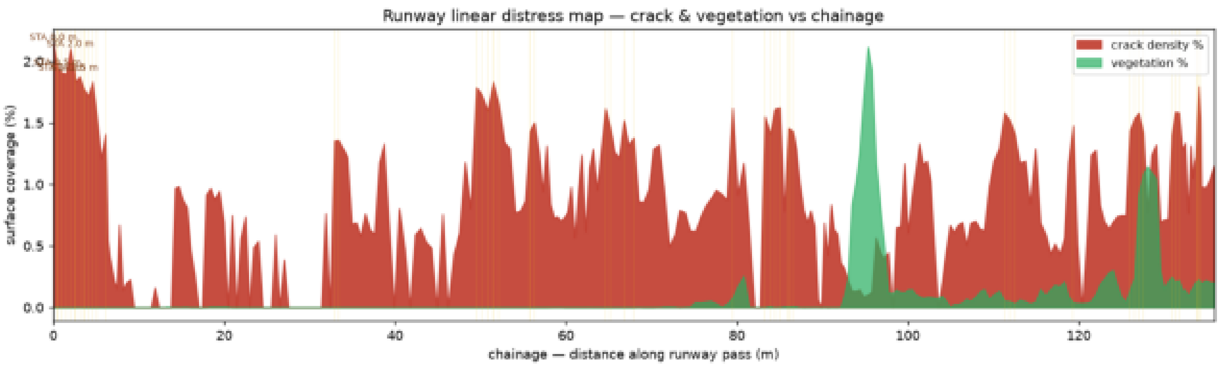

Figure 5. Crack and vegetation coverage plotted against chainage — distance along the pass in metres. The concentration of cracking in the first few metres marks the priority stretch; vegetation rises toward the far end.

A list of defects is only useful if each one can be found again on the ground. With no satellite positioning recorded, we recover position from the imagery itself: consecutive frames overlap heavily, so aligning each frame with the next gives the distance the platform moved between them, which accumulates into a chainage — a running distance along the runway. The platform advanced about half a metre per frame, and the surveyed length comes to 136 m. Every crack and patch of vegetation then carries a station value, and the survey reads as a map rather than a list (Figure 5).

6. Results

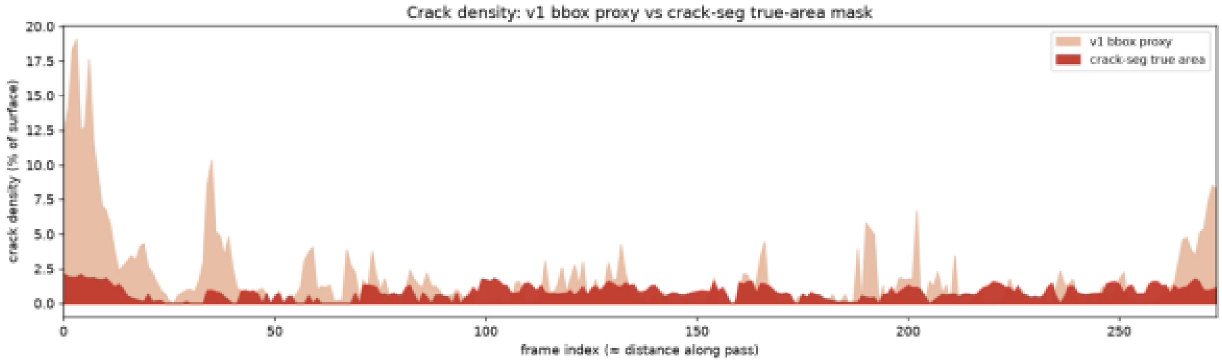

Figure 6. Crack density along the runway, comparing a coarse bounding-box estimate with the true area from segmentation. The outline measurement corrects the magnitude while preserving the pattern of where cracking concentrates.

Crack density — the share of surface taken up by cracking — is the headline figure for prioritisation. Measuring it from outlines rather than boxes matters: a box drawn round a thin crack vastly overstates the affected area, whereas the outline gives the true figure of about 0.82% of surface (Figure 6). Both views agree on where the trouble is — the heaviest cracking sits in the first few metres of the pass — but only the outline gives a defensible number. The vegetation track adds a second, independent map of where the verge is breaking down. Together they let a maintenance team see, at a glance, which stretches to walk first.

7. Limitations

We are deliberate about what this survey does not claim. We do not publish a per-crack width, an ASTM D5340 severity grade, or a single condition index, because the imagery cannot support them honestly. The grooved surface and the motion blur of a moving platform both blur the thin edge of a crack — exactly the edge a width measurement depends on — and across three independent attempts the width either inflated to physically impossible values or collapsed onto the spacing of the grooves rather than the cracks. Since width drives severity and the index, we leave them out rather than report a number we cannot stand behind. Two further caveats: formal accuracy figures would need a human to confirm a sample of frames, and a single straight pass yields a strip down the runway rather than a full-width map of the whole surface.

8. Conclusions and outlook

From one short drone pass, the approach delivers a located, prioritised read of a runway's surface: where the cracking is and how it runs, where vegetation is taking hold, and which stretches deserve the first walk — all tied to a real distance along the runway and framed against airfield practice. The clear next steps follow directly from the limitations. Sharper capture — a slower or stop-and-stare pass that removes the motion blur — would unlock trustworthy crack width and, with it, a defensible severity grade and condition index. A crack model fine-tuned on this specific surface would sharpen the outlines further, and a second pass offset across the runway would extend the strip into a full-width map. None of these change the architecture; each simply strengthens a track that is already in place.

References

[1] ICAO. Annex 14 to the Convention on International Civil Aviation — Aerodromes.

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[3] ASTM D5340, Standard Test Method for Airport Pavement Condition Index Surveys.

[4] FAA AC 150/5320-17A, Airfield Pavement Surface Evaluation and Rating.

[5] Li et al., Automated quantification of crack length and width in asphalt pavements, Computer-Aided Civil and Infrastructure Engineering, 2024.

[6] OpenSistemas, YOLOv8 crack-segmentation models, Hugging Face.

[7] Reddy et al., Real-time 2D orthomosaic mapping from drone imagery via sequential image registration, 2023.